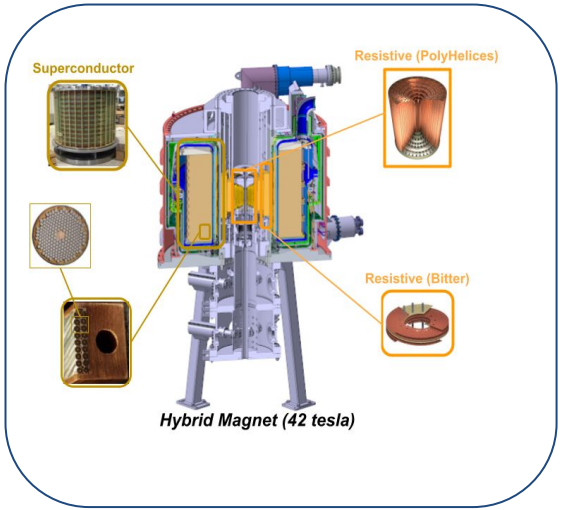
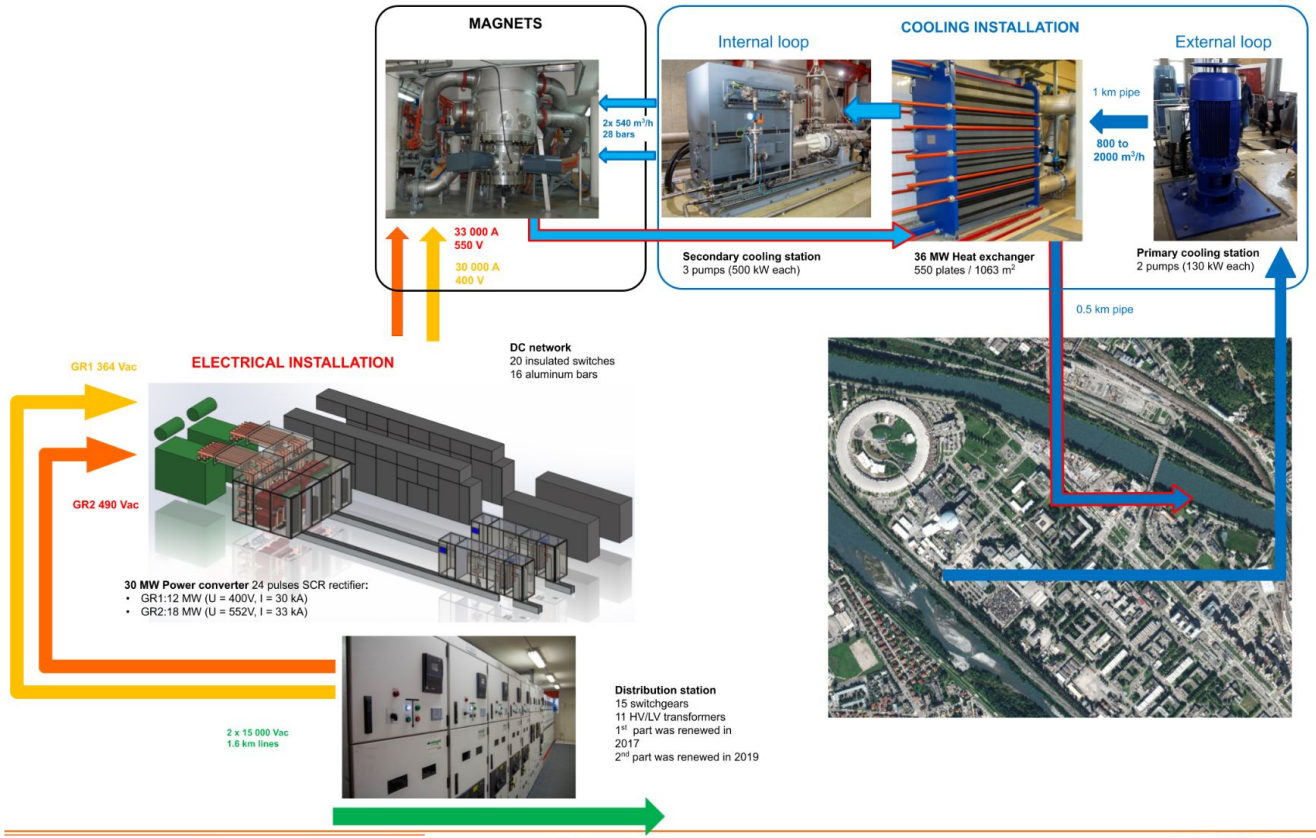
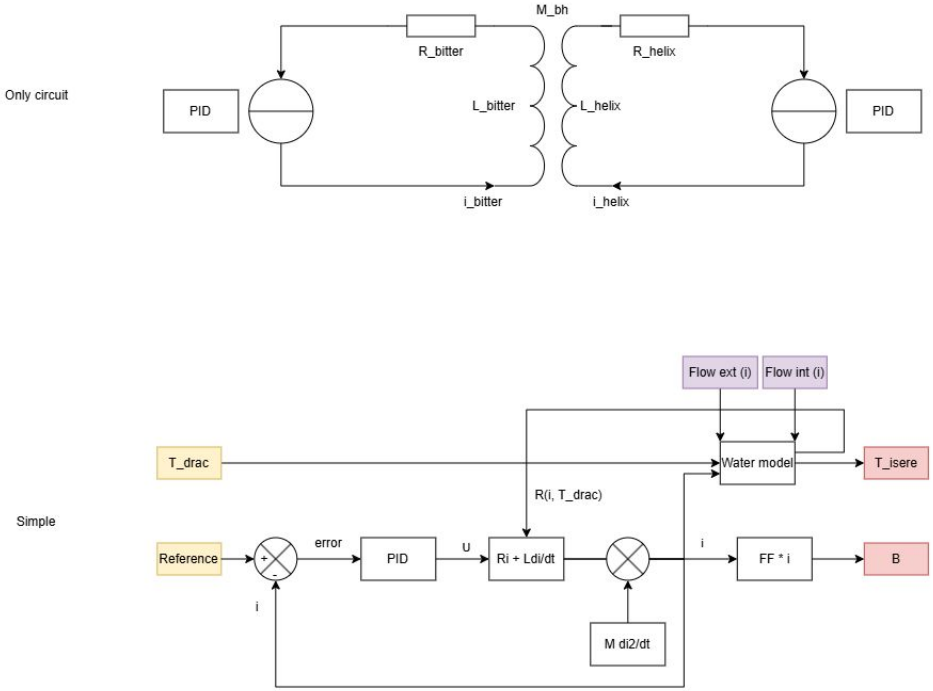
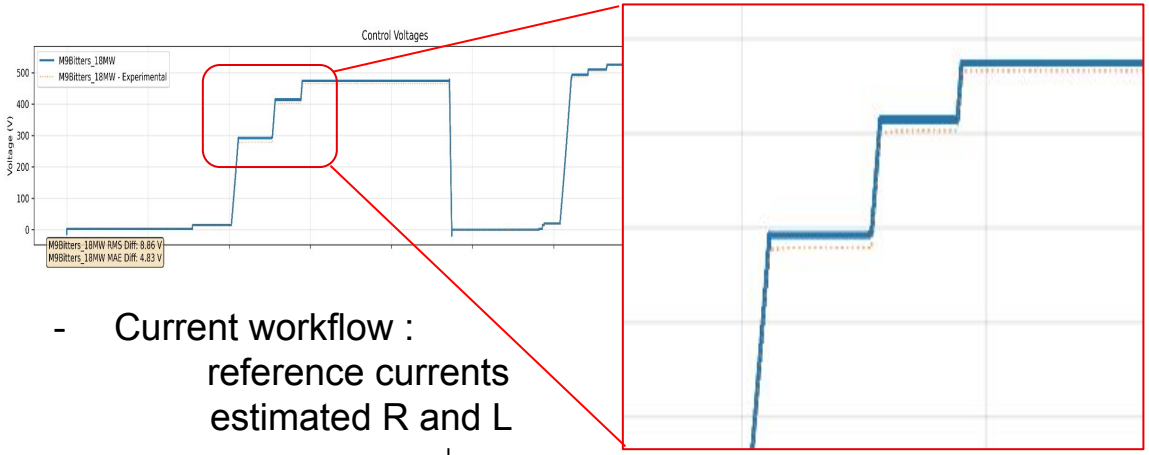
LNCMI-G: A Magnetic Field User Facility





- ***Numerical Commissioning (feelpp)***:
  - R(I), Inductances
  - Store data in specific table(s) in DB
- ***Magnet in operation***:
  - ODEs electric circuits
  - + PID
    - ➢ Get params from DB
  - + Cooling system
    - ➢ Flow rate control
    - ➢ Heat exchanger model





- Current workflow :
  reference currents
  estimated R and L
  ↓
  ODE solver
  ↓
  times series outputs
  (Tw, Voltage, Power, Current)

- simulation time : ~4.4 h
- computation time : ~ 1 hour (sequential on Intel Xeon 2.4 Ghz)

Current implementation :

- config: json file
  - input:
    - use magnetrun to csv
    - geometry: yaml config
    - resistances/inductances
  - python package
    - scipy ODE solver
    - Dorma/Price algo
  - resultats:
    - csv from operational data

- ***Cooling Loops***:
  - Use to model transients: inputs (temperatures from cold/hot side, Flow rate)
  - Comparisons with output temperature (both cold /hot side)
  - Trials to improve predictions

